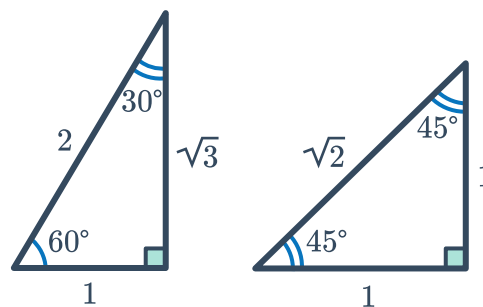


## Exact value triangles



1 Find the exact value of the following:

- |                   |                   |
|-------------------|-------------------|
| a $\sin 30^\circ$ | b $\sin 60^\circ$ |
| c $\sin 45^\circ$ | d $\cos 45^\circ$ |
| e $\cos 60^\circ$ | f $\cos 30^\circ$ |
| g $\tan 60^\circ$ | h $\tan 45^\circ$ |
| i $\tan 30^\circ$ |                   |



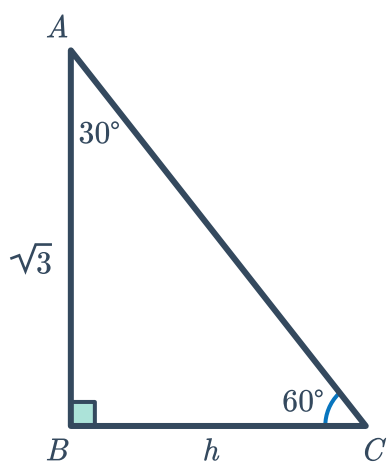
2 Evaluate the following expressions, leaving your answer in exact, rationalised form:

- |                                         |                                                 |
|-----------------------------------------|-------------------------------------------------|
| a $\frac{\sin 60^\circ}{\cos 30^\circ}$ | b $\sin 45^\circ \cos 30^\circ$                 |
| c $\sin 45^\circ + \cos 30^\circ$       | d $\sin 45^\circ \cos 60^\circ + \tan 45^\circ$ |

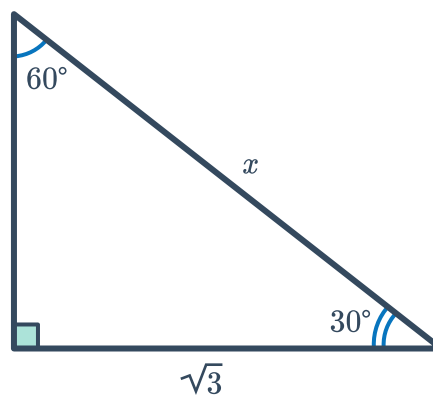
## Unknown sides

3 Find the exact value of the pronumeral in the following triangles:

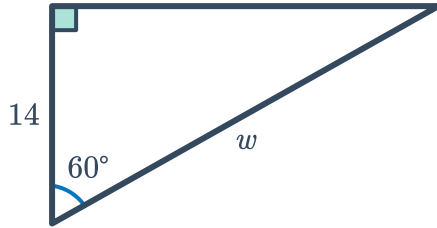
a



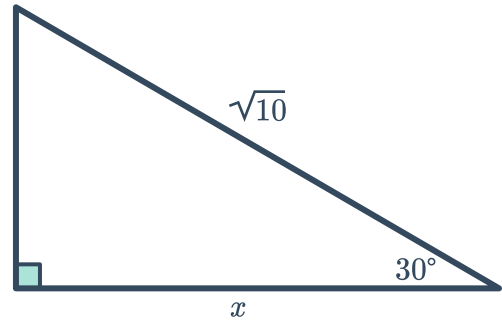
b



c

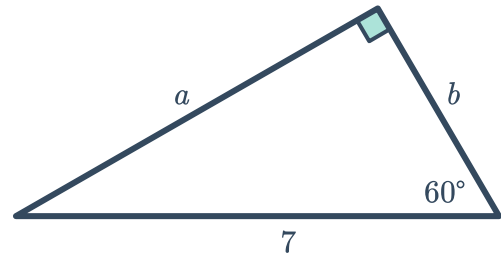


d



4 Consider the given triangle:

- a Find the exact length of side  $a$ .
- b Find the exact length of side  $b$ .

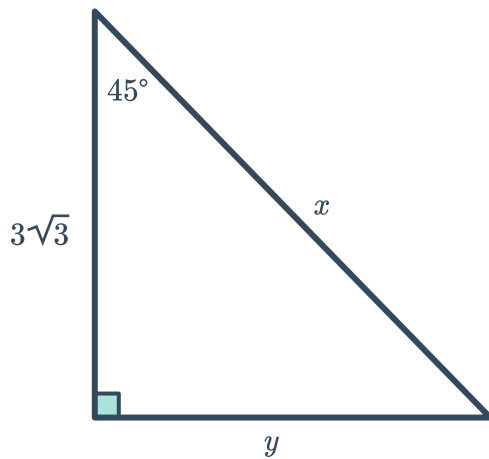


5 For each of the following triangles:

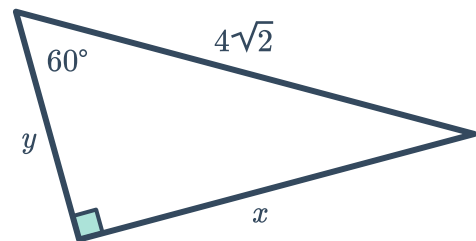
i Find the exact value of  $x$ .

ii Find the exact value of  $y$ .

a

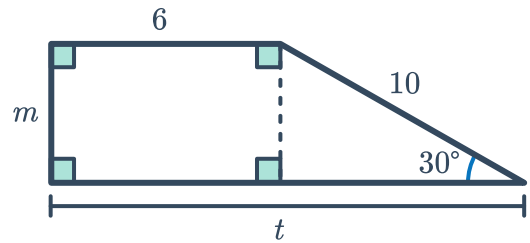


b



6 Consider the unknown lengths in the given figure:

- a Find  $m$ .
- b Find  $t$ .



## Unknown angles

7  $\theta$  is an angle in a right-angled triangle. Find the value of  $\theta$  in the following equations:

a  $\cos \theta = \frac{1}{2}$

b  $\sin \theta = \frac{\sqrt{3}}{2}$

c  $\sin \theta = \frac{1}{\sqrt{2}}$

d  $\tan \theta = \frac{1}{\sqrt{3}}$

8 Find the value of  $\theta$  if  $\cos \theta = \frac{1}{\sqrt{2}}$  and  $\sin \theta = \frac{1}{\sqrt{2}}$ .

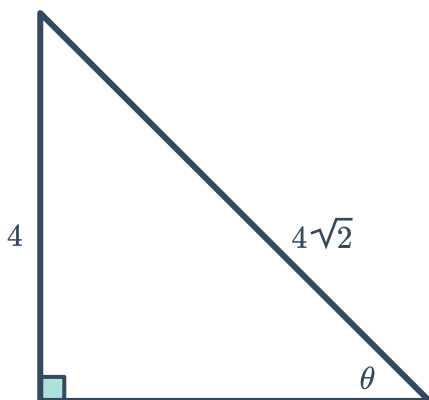
9 Given that  $\cos \theta = \frac{\sqrt{3}}{2}$  and  $\sin \theta = \frac{1}{2}$ :

a Find the value of  $\theta$ .

b Find the value of  $\tan \theta$ .

10 Find the unknown  $\theta$  in the following triangles:

a



b

